

Thermal Conductivity of

Selected Materials

part 1

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A. Introduction

The nonmetallic solids treated in this part include aluminum oxide, beryllium oxide, Corning code 7740 glass, diamond, magnesium oxide, Pyroceram brand glass-ceramic code 9606, quartz, thorium dioxide, and titanium dioxide. The thermal conductivities of these solids cover a fairly wide range.

A thorough literature search was conducted for the thermal conductivities of these solids. The available experimental data are collected and shown in Figures 15 to 23. Each figure is followed by a specification table giving information on the specimens corresponding to the respective curves in the figure. For curves listed in the specification table but not shown in the figure, particularly in the low temperature region, one is referred to the TPRC Data Book Volume III, Chapter 1 [1], in which complete numerical data tables are also given. The heavy broken line shown in the figure represents the recommended curve of thermal conductivity versus temperature. For each of the oxides, except quartz, the recommended curve is for a 99.5% pure, 98% dense polycrystalline specimen. In all the oxide single crystals, only the data for quartz single crystals are analyzed at this time.

The recommended values are the results of critical analysis of available data from all sources. The analysis procedure involves careful evaluation of the validity of available data and related information, resolution and reconciliation of the disagreements of conflicting data, correlation of data with affecting parameters, comparison of the resulting values with theoretical predictions and so forth.

According to theory, the thermal conductivity of nonmetallic crystals is determined both by the specific heat and the mean free path of the phonons. At temperatures close to absolute zero the phonons are scattered by the boundaries; the thermal conductivity varies as T^3 and is size dependent. As the temperature is increased, other scattering mechanisms become effective: scattering of phonons by static imperfections (impurities, isotopes and all kinds of defects) and scattering of phonons by other phonons (Umklapp processes). The last mentioned process increases rapidly with temperature, until the mean free path decreases more rapidly with temperature than the specific heat increases. At this point, still at very low temperatures (usually below $\theta/20$), the thermal conductivity passes through a maximum, and then decreases again, in some perfect crystals exponentially. Around the Debye temperature and above, it should vary as T^{-1} .

However, experimental thermal conductivity data measured at high temperatures show numerous exceptions to this T^{-1} rule. In some cases the thermal conductivity varies more slowly than T^{-1} . This can be ascribed to scattering by imperfections. At high temperatures the resistance due to Umklapp processes varies as T , that due to imperfections is more or less independent of temperature, and the combined resistance has an intermediate variation, hence the conductivity varies more slowly than T^{-1} . This is particularly pronounced in mixed crystals and in highly disordered crystals. Furthermore, in the extreme case of disorder such as in glasses and in vitreous materials, the disorder determines the phonon mean free path which consequently becomes constant at high temperatures. Hence the thermal conductivity increases with temperature, being roughly proportional to the specific heat of the material.

Other cases of departure from the theoretical behavior may be explained in terms of a radiative component of heat transfer. In porous materials, a radiative heat transfer across the pores acts as a shunt. In some cases the crystals are at least partly transparent to infrared radiation, and there is an internal radiation component. In either case the additional thermal conductivity is proportional to the specific heat of the radiation component (proportional to T^3) and the mean free path of the photons. In some cases this mean free path is governed by internal absorption and scattering processes; the radiation component is then an intrinsic property of the material. In other cases the material is transparent, and the photons traverse the specimen. The corresponding component of heat transfer is thus dependent on the dimensions of the specimen, analogous to the case of phonon heat transport near absolute zero. Finally, it may happen that photons move from the heat source in the measuring apparatus to the heat sink, partly going through the specimen and partly through free space. In this case it seems diffi-

cult to separate radiative measurement error from internal radiative heat transport. To compound the conceptual difficulty, since the internal absorption and scattering probabilities depend on photon frequency, it may happen that part of the radiative conductivity is intrinsic and part dependent on specimen dimensions. To sum up, the radiative component of the thermal conductivity can be extremely complex, is generally not well understood, and often increases rapidly with temperature, though in principle it may also decrease over some range of temperatures, depending on the photon absorption coefficient.

In analysing the available data, attempts were made to consider all parameters likely to affect the thermal conductivity, such as purity, crystal imperfection, crystal axes orientation, density (or porosity), thermal history, microstructure, etc. However, it is very unfortunate that in the majority of cases the authors do not report all necessary pertinent information to fully characterize the materials for which their data are reported. Data for unspecified materials are of little utility. The recommended values have therefore at times been based on scanty information and should be accepted as an attempt to make the most of existing knowledge and subject to modification in the light of further work.

For a material to be used as a reference standard, the chemical and physical stability is a major requirement. The stability of a material is very much dependent on the environment in which heating to high temperatures takes place. For instance, the stoichiometric composition of titanium dioxide is affected by heating to high temperatures in reducing environment and consequently its thermal conductivity is likely to change after such heating. Therefore, selection of a suitable environment is important in using a reference material.

Some pertinent comments regarding the thermal conductivity of each solid are given in the following section, and the recommended values are reported collectively in Table 14 and in Figure 14. In the table, the third significant figure is given only for the purpose of comparison and for smoothness and is not indicative of the degree of accuracy. The recommended values for most of these materials do not cover the low temperature region. This is because either there are no experimental data available in the low temperature region for some of the materials or the pertinent information on specimen characterization is inadequate. It is noted that in the low temperature region the thermal conductivity of a crystal is very strongly dependent on its purity and perfection, among other parameters, and therefore knowledge of the sample characterization is particularly important at low temperatures.

B. Thermal Conductivity of A Group of Selected Nonmetallic Solids

Aluminum Oxide. - There are 169 sets of experimental thermal conductivity data available for aluminum oxide, of which 34 sets are for single crystals and the remaining 135 sets are mostly for polycrystalline specimens and for some alumina powder. Despite these large numbers, the specimen purity and density (or porosity) are known for only one set of data for the single crystals, and for only 25 sets of data for the polycrystalline specimens. It is indeed a waste of effort to make such extensive measurements on specimens whose major specification and characterization are not known.

The available data for polycrystalline specimens cover the temperature range from 2.5 to 2030 K. There are four sets of data below room temperature. One set of data (curve 130) bridges the gap and covers the low, normal, and moderately high temperature regions. However, the latter appeared to be not of the correct trend. All the remaining data are above room temperature. They spread into a wide band, as shown in Figure 15, over the entire temperature range where data are available. The spreading of the curves is mainly due to the differences in purity and density of the specimens measured, and therefore effectively each curve represents a different material. It should be noted here that, because of the nature of these different curves, an average curve obtained through some form of statistical fit of all these data points would have little meaning.

From the inadequate information available on specimen specification and characterization, the analysis of thermal conductivity data can be based only on porosity and purity. The thermal conductivity values of various curves at each of several selected temperatures were read off a large working-graph, and were used to determine

the relationship between thermal conductivity versus porosity and purity by means of an iteration technique. The procedure used was as follows.

The thermal conductivity values were first plotted against porosity of the specimens neglecting at first the differences in specimen purity. A preliminary curve was thus obtained at each temperature representing the tentative relationship between thermal conductivity and porosity. Using this preliminary result the thermal conductivity values were adjusted to zero porosity, and the resulting zero-porosity thermal conductivity values were plotted against specimen purity. Thus a second generation preliminary curve was obtained at each temperature representing the tentative relationship between thermal conductivity and purity. Again using this second generation preliminary result the thermal conductivity values were adjusted to zero impurity, and the resulting zero-impurity thermal conductivity values were plotted against porosity. By repeating this procedure, two final curves were obtained representing the relationships between thermal conductivity versus purity and porosity. It must be pointed out that the resulting relationships are still very preliminary in nature due to the inadequacy of available information.

The relationship between thermal conductivity and porosity derived from experimental data is found in closer agreement with the relationship derived by Euler [2] from the theoretical model of Maxwell [3] than with the simplified relation derived by Loeb [4, 5].

Using the above correlation procedure, at each of the several selected temperatures, a point was obtained corresponding to the thermal conductivity of a 99.5% pure, 98% dense polycrystalline specimen. A mean curve was subsequently drawn through these points to serve as the recommended curve.

The recommended curve has a minimum at about 1700 K, in agreement with the general trend of the experimental data. Above 1700 K it turns up probably mainly due to the increasing contribution of the radiation component. The uncertainty of the recommended values should be within $\pm 8\%$ at temperatures from 500 to 1000 K and increases to about $\pm 15\%$ below 250 K and above 1800 K.

Beryllium Oxide. - The thermal conductivity of beryllium oxide is exceptionally high, in fact, higher than that of most refractory materials. At temperatures below about 500 K it exceeds the thermal conductivity of beryllium.

There are 58 sets of data available for polycrystalline beryllia. No data are available for single crystals. Four sets of data are below room temperature and the rest are above. Out of the 58 sets of available data, only 15 sets are for specimens of known purity and density (or porosity). The purity and density of most specimens are low.

In the measurements of the thermal conductivity of five specimens of beryllium oxide of different density, Powell [6] found that, within the temperature range studied (325 to 825 K), the thermal conductivity of beryllia varies approximately inversely as the absolute temperature and, at any one temperature, varies approximately linearly with the density.

The thermal conductivity data were correlated with porosity and purity. Several points corresponding to a 99.5% pure, 98% dense polycrystalline specimen were obtained at several selected temperatures. The recommended curve, as shown in Figure 16, has been drawn through these points. It has a minimum at about 2100 K and turns up above this temperature. This trend of the thermal conductivity curve has been investigated and confirmed experimentally by several workers including Taylor [7].

The uncertainty of the recommended values is thought to be within $\pm 8\%$ at temperatures from 500 to 1000 K and increases to $\pm 15\%$ below 300 K and above 1800 K. Beryllium oxide has a phase transformation at about 2370 K. This may limit the use of beryllia as a high-temperature thermal conductivity standard.

Corning Code 7740 Glass. - This glass is produced by Corning Glass Works and is a particular kind of borosilicate glass, commercially known as Pyrex glass. It is composed approximately of 80.6% SiO_2 , 13% B_2O_3 , 4.3% Na_2O , and 2.1% Al_2O_3 , which is the average of the compositions reported for four of the specimens measured.

The spreading of the thermal conductivity curves as shown in Figure 17 is probably mainly due to experimental errors since small variations of chemical composition or microstructure in different specimens of this glass are unlikely to cause very large differences in thermal conductivity*.

There are 12 sets of thermal conductivity data available for this glass. The specimen measured by Birch and Clark [8] (Curve 10) was stated as No. 774 Pyrex glass from Corning Glass Works instead of No. 7740, but was actually the same kind of glass, since No. 774 was the old designation. The data of curves 2 and 6, which spread out the most, were derived from thermal diffusivity measurements and appear to be in error. The recommended curve has been drawn through the mean of the remaining 10 curves.

The recommendations are thought to be accurate to within $\pm 5\%$ at moderate temperatures and within $\pm 10\%$ below 200 K and above 700 K.

Diamond. - Despite its high cost, diamond is, from a theoretical standpoint, the obvious choice for a reference standard due to the fact that it forms a monatomic, isotropic crystal lattice, reasonably free from defects and most close to the simple crystalline model on which the theory of thermal conductivity is based. Furthermore, diamond retains rather well its geometry and degree of perfection. At room and moderate temperatures diamond has the highest thermal conductivity of all materials.

The water-white diamonds are divided into types I and II, according to whether the ultraviolet absorption by the diamond is pronounced at wavelengths near 3000 Å or 2200 Å, respectively [9, 10]. Type II diamonds are in turn subdivided into two kinds: type IIa which refers to the insulating variety and type IIb to the good conductors of electricity.

There are 14 sets of data available for diamond as shown in Figure 18. It can be seen that only for type I does more than one experimental curve exist. Moreover, most of the measurements have come from the Clarendon Laboratory, Oxford [11, 12], and for type I the workers concerned regarded the newer measurements (curve 14) as being more accurate. Therefore, there is no comparison or choice that can be made from the existing data for diamond. It is very tentatively recommended that the values represented by the three smooth curves for the three types of diamond, in the temperature range from 150 to 400 K, be provisionally accepted as most probable values pending the availability of further experimental data. Until further data are available an attempt to estimate the accuracy of the present values does not seem justified.

Magnesium Oxide. - Fifty-two sets of experimental data are available, in which six sets are for single crystals and the remaining 46 sets are mostly for polycrystalline specimens and for some magnesia powder. All data available for polycrystalline specimens are at temperatures above room temperature. Out of the 52 sets of available data, only 13 sets are for specimens of known purity and density (or porosity). The purity and density of most specimens are low.

Charvat and Kingery [5] have investigated the effect of purity, porosity, and microstructure on the thermal conductivity of magnesium oxide among other materials. They give fairly complete information on the specification and characterization of their specimens, which many other workers fail to do. However, their data appear to be high and the inclusion of their data in the correlation of thermal conductivity with purity and porosity would probably lead to high results.

The recommended curve as shown in Figure 19 is for a 99.5% pure, 98% dense polycrystalline specimen, and is the result of the correlation of thermal conductivity with purity and porosity. This curve has a minimum at about 1650 K and turns up at higher temperatures. The uncertainty of the recommended values is thought to be within $\pm 8\%$ at temperatures from 500 to 1000 K and increases to $\pm 15\%$ below 300 K and from 1650 to 2000 K. The uncertainty increases to about $\pm 20\%$ above 2100 K.

*Dr. P. G. Klemens (private communication) suggests that even at ordinary temperatures the thermal conductivity of glass is governed by the mean free path of relatively long wave phonons, and that this mean free path may be sensitive to the occurrence of very small crystallites. This may be a greater source of variability in the thermal conductivity than chemical composition.

Pyroceram Brand Glass-Ceramic Code 9606. - This is a microcrystalline material composed of silicon dioxide, aluminum oxide, magnesium oxide, and a small amount of titanium dioxide, and is produced by Corning Glass Works.

There are eight sets of experimental data available. As shown in Figure 20, six curves are in the temperature range from 93 to 1433 K. Of these six curves only two (curves 1 and 2) were obtained by direct measurements of thermal conductivity. The other four curves were derived from thermal diffusivity data. In view of the fact that these curves were obtained by completely different methods, the agreement of the data with one another is rather good.

Curves 1 and 2 were produced by Robinson and Flynn [13] and their experimental data are usually reliable. The recommended curve has been drawn closely following these two curves. At temperatures above 500 K the recommended curve deviates slightly from curve 1 to follow the general trend of all the curves, since at temperatures above 600 K curve 1 is essentially a straight line which is deemed not very likely. Flieger [14] observed a small jump in his thermal diffusivity data. This jump is ignored at this time pending further confirmation.

The recommended values should be accurate to within $\pm 5\%$ at temperatures from 200 to 1000 K. The uncertainty increases to $\pm 10\%$ at 100 K and 1400 K.

Quartz. - There are 45 sets of experimental thermal conductivity data available for quartz single crystals, of which 13 sets, 22 sets, and one set are, respectively, for single crystals measured with the heat flow direction parallel, perpendicular, and at 45 degrees to the c-axis of the crystal, and nine sets for single crystals of unknown axis orientation. Four sets of data are available for quartz powder and 78 sets of data available for fused quartz, which is called by various commercial names such as fused quartz, quartz glass, fused silica, vitreous silica, silica glass, etc. It is surprising to note that out of these 127 sets of data only one set is for a specimen whose purity and density are both known. The specimen purity is known for only two other curves and the specimen density is known for only 28 other curves.

The available data and pertinent information were studied, critically evaluated, and selected. The recommended curve as shown in Figure 21 for a high-purity (99.99%) quartz single crystal measured with the heat flow direction parallel to the c-axis has been drawn as the mean of curves 39, 14, 49, 51, 24, and 54. The recommended curve for a high-purity (99.99%) quartz single crystal measured with the heat flow direction perpendicular to the c-axis has been drawn as the mean of curves 6, 15, 50, 53, 23, 17, 3, 2, and 55. The recommended curve for high-purity (99.99%) clear fused quartz has been drawn as the mean of curves 45, 46, 47, 48, 110, 12, 13, 11, 16, 111, 120, 66, 67, 68, 30, 84, and 29. The above curve numbers are listed in such an order that the curves commence at increasing temperatures.

Quartz crystal has a phase transformation which occurs at about 848 K. Experimental specific heat data [15, 16, 17] and thermal expansion data [18] indicate a pronounced discontinuity at the transition temperature. The effect of this phase transformation on the thermal conductivity of quartz crystal is not known, since no measurement has ever been made above 750 K.

The uncertainty of the recommended values for quartz single crystal is thought to be within $\pm 5\%$ at temperatures from 300 to 500 K and increases at lower and higher temperatures up to $\pm 10\%$ at 40 and 800 K. The uncertainty of the recommended values for fused quartz should be within $\pm 4\%$ at temperatures from 200 to 500 K, and increases to $\pm 8\%$ at 50 K and 900 K and up to $\pm 15\%$ below 10 K and at 1400 K.

Thorium Dioxide. - There are only six sets of experimental thermal conductivity data available for thorium dioxide over the temperature range from 304 to 379 K and from 527 to 1821 K. No data are available below room temperature and between 379 and 527 K. Furthermore, all the available data are for polycrystalline specimens and no measurements have yet been made on thoria single crystal.

From the available inadequate information, the specimen purity and density are known for only one set of data, and density is known for the other curves. Below 380 K the only two available curves, 1 and 2, were

produced by the same author on the same specimen. The difference in the experimental results was attributed by the author to the difference in the coatings used on the end faces of the cylindrical specimen in separate runs, and he pointed out that the data obtained during the second run (curve 2) using a platinum alloy glaze on the faces of the cylinder were more accurate than those obtained during the first run (curve 1) in which a gold coating was used. The recommended values at the lower temperatures are therefore based on curve 2.

Several points, each at one of the several selected temperatures, were derived from the available meager information to correspond to a 99.5% pure, 98% dense polycrystalline specimen, and the recommended curve has been drawn through these points. The uncertainty of the recommended values is thought to be within $\pm 15\%$ at temperatures from 350 to 1000 K and increases at lower and higher temperatures up to $\pm 20\%$ at 250 K and $\pm 25\%$ at 1900 K.

Titanium Dioxide. - There are 36 sets of experimental thermal conductivity data available for titanium dioxide, of which 21 sets are for single crystals, 13 sets for polycrystalline specimens and 2 sets are for an unspecified specimen measured before and after irradiation. The specimen purity and density (or porosity) are both known for only six sets of data.

For the polycrystalline titania all available data are at temperatures above 350 K, and it is interesting to note that all curves except two were produced by workers of the same institution. Their later results [5] were 10% to 50% higher than earlier ones, which was attributed partially to the use of dense high-purity specimens and partially to the improved design of the thermal conductivity apparatus used in their later measurements. However, on comparison of their results with the recently published data for extremely high-purity titania single crystals [19], their later results [5] appear to be somewhat too high. On account of this fact, adjustments and modifications have been made in deriving the recommended values from their data.

The recommended values are for a 99.5% pure, 98% dense polycrystalline specimen. The uncertainty of the recommended values should be within $\pm 10\%$ at temperatures from 400 to 1000 K and increases to $\pm 15\%$ at 250 K and 1400 K.

TABLE 14. RECOMMENDED THERMAL CONDUCTIVITY OF NONMETALLIC SOLIDS*

Thermal Conductivity, k, Watt cm⁻¹ K⁻¹

T, K	Al ₂ O ₃ **	BeO**	Corning Code 7740 Glass	Diamond†		Pyroceram Brand Glass-Ceramic Code 9606	Quartz† (Single Crystal)		Quartz† (Clear Fused)	ThO ₂ **	TiO ₂ **	T, K
				Type I	Type IIa Type IIb		to c-axis	⊥ to c-axis				
0	0	0	0	0	0	0	0	0	0	0	0	0
1									(0.00024)			1
2									(0.00054)			2
3									0.00080			3
5									0.00118			5
10									0.00127			10
20									0.00153			20
40									0.00266			40
60							1.79	0.89	0.0041			60
80							0.85	0.429	0.0055			80
100	1.33		0.0058	30	100	0.0542	0.54	0.279	0.0069			100
150	0.77		0.0076	20	60.5	0.0550	0.39	0.208	0.0095			150
200	0.55	4.24	0.0090	14.2	40.4	0.0474	0.231	0.130	0.0114			200
250	0.434	3.34	0.0101	11.0	29.7	0.0428	0.164	0.095	0.0128	(0.152)	(0.093)	250
273	0.397	3.02	0.0106	10.0	26.5	0.0413	0.127	0.075	0.0133	(0.142)	(0.089)	273
300	0.360	2.72	0.0110	9.0	23.5	0.0399	0.116	0.0684	0.0138	(0.132)	(0.084)	300
350	0.307	2.28	0.0117	(7.6)	(19.5)	0.0379	0.104	0.0621	0.0145	0.115	(0.0767)	350
400	0.264	1.96	0.0124	(6.6)	(16.6)	0.0365	0.088	0.0530	0.0151	0.102	0.0701	400
500	0.202	1.46	0.0136			0.0345	0.076	0.0470	0.0162	0.081	0.0588	500
600	0.158	1.11	0.0149			0.0331	0.060	0.0388	0.0175	0.066	0.0502	600
700	0.126	0.87	0.0165			0.0319	0.050	0.0340	0.0192	0.055	0.0439	700
800	0.104	0.70	(0.0189)			0.0310	0.047	0.0314	0.0217	0.047	0.0394	800
900	0.089	0.57				0.0303	(0.0420)	(0.0306)	0.0248	0.041	0.0365	900
1000	0.0785	0.47				0.0297			0.0287	0.0368	0.0346	1000
1100	0.0710	0.39				0.0291			0.0336	0.0336	0.0335	1100
1200	0.0655	0.33				0.0287			0.0400	0.0312	0.0328	1200
1300	0.0613	0.283				0.0284			0.0482	0.0296	0.0323	1300
1400	0.0585	0.245				0.0282			0.0620	0.0284	0.0321	1400
1500	0.0566	0.215								0.0273		1500
1600	0.0556	0.195								0.0266		1600
1700	0.0554	0.180								0.0259		1700
1800	0.0559	0.167								0.0254		1800
1900	0.0574	0.156								(0.0252)		1900
2000	0.0600	0.150										2000
2100	(0.0644) †	0.150										2100
2200		0.152										2200
2300		0.164										2300

*In the table the third significant figure is given only for the purpose of comparison and for smoothness and is not indicative of the degree of accuracy.

**99.5% pure, 98% dense, polycrystalline.

†High-purity.

‡Values in parentheses are extrapolated.